

FedMKS: A Hybrid Model and Knowledge Sharing Approach for Robust Federated Learning in Heterogeneous Networks

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Abstract—Federated Learning (FL) enables distributed clients to train a shared model without exchanging raw data, making it attractive for privacy-sensitive edge applications. Real deployments, however, face *system heterogeneity*: devices differ widely in compute capability, so no single model architecture fits all clients. Knowledge-distillation-based FL sidesteps this constraint by exchanging soft labels on a shared public set instead of weights, allowing each client to keep its own architecture. Yet under *statistical heterogeneity* (non-IID label skew), soft-label-only distillation degrades sharply and can underperform simple weight averaging on a common small model. We present FedMKS (Federated Learning with Model and Knowledge Sharing), a hybrid algorithm that organises clients into architecture tiers and couples intra-tier weight averaging with cross-tier soft-label distillation, recovering the accuracy of weight sharing while preserving architectural diversity. We apply FedMKS to indoor Wi-Fi CSI sensing for occupancy and activity detection—a privacy-critical setting where raw data cannot leave the home—and evaluate it on the HomeOccupancy CSI dataset and MNIST/CIFAR-10 under IID and non-IID partitions against FedMD, FedAKD, FedAvg, FedProx, FedDF, and Local/Centralised baselines. FedMKS attains 90.4% on HomeOccupancy IID—within 0.5% of the centralised ceiling. Code and datasets are publicly available.

I. INTRODUCTION

Federated Learning (FL) allows a population of edge devices to collaboratively train a model while keeping raw data on device [1]. This privacy-preserving paradigm is appealing wherever data is sensitive, scarce, or expensive to centralise—smart homes, healthcare, and mobile sensing being canonical examples.

A persistent obstacle is *system heterogeneity*: edge devices range from constrained sensors to capable gateways, so no single architecture fits all. Weight-sharing methods (FedAvg, FedProx) require identical architectures and therefore force every client onto the smallest common model. Knowledge-distillation methods (FedMD, FedAKD, FedDF) remove this constraint by exchanging soft labels on a shared public set rather than weights, letting each client keep an architecture matched to its hardware.

Distillation alone, however, suffers under *statistical heterogeneity*: when client label distributions are non-IID, soft-label averaging provides a weaker training signal than weight

averaging, and our experiments show KD-only methods trailing even all-small weight sharing by double-digit margins. This motivates a *hybrid* architecture that combines both channels—averaging weights where architectures match and distilling soft labels where they do not.

We instantiate this idea for indoor Wi-Fi sensing: Channel State Information (CSI) encodes fine-grained multipath propagation effects that shift when a person moves, breathes, or changes posture, enabling camera-free occupancy and activity recognition from commodity access points [2], [3]. Labelled CSI is private and hard to collect at scale, making it a natural application for heterogeneous FL.

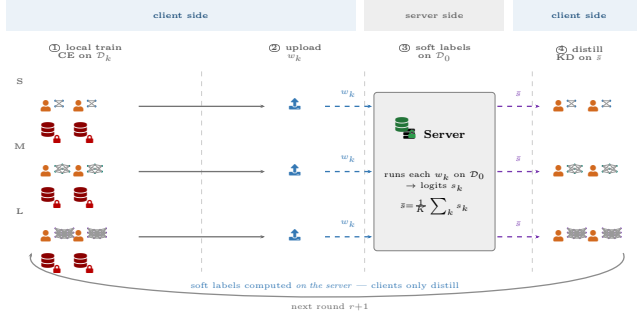
Contribution. We propose **FedMKS** (Federated Learning with Model and Knowledge Sharing), which (i) partitions clients into architecture tiers and shares classifier weights within each tier, (ii) shares distilled soft labels across tiers on a public set, and (iii) applies Mixup augmentation before distillation. A dual-head tiered model decouples private classification from public-set distillation, so clients of different capacity collaborate without exchanging raw data.

II. RELATED WORK

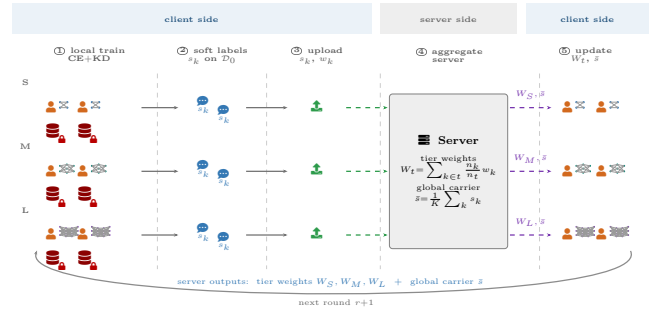
Figure 2 sketches the FL research timeline relevant to this work; we review federated learning in general, distillation-based FL for system heterogeneity, the impact of statistical heterogeneity, and Wi-Fi sensing as our target application.

A. Federated Learning

FedAvg [1] introduced federated averaging of locally trained weights and remains the canonical FL baseline, but it requires identical architectures across clients; FedProx [5] adds a proximal term that tolerates statistical and system heterogeneity while keeping the same architectural constraint. Sozinov et al. [6] first showed federated HAR can rival centralised accuracy while preserving privacy; Wu et al. [7] extended FL to CSI sensing, where non-IID per-home distributions degrade FedAvg and motivate personalised approaches [8]–[10].



(a) FedDF [4]: soft labels computed *on the server*.



(b) FedMKS (ours): soft labels computed *on each client*; server also averages intra-tier weights.

Fig. 1. **FedDF vs. FedMKS — one communication round.** FedDF uploads client weights and runs every model on the public set at the server; FedMKS computes soft labels s_k on each client, so the server only averages tensors and additionally FedAves intra-tier weights (W_S, W_M, W_L).

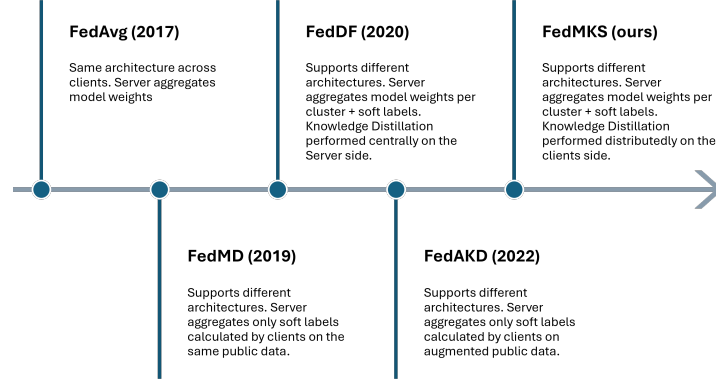


Fig. 2. Timeline of federated-learning research relevant to this work.

B. Knowledge-Distillation-Based FL

FedMD [11] removes the architecture constraint by sharing only soft predictions on a public set, building on Hinton’s distillation framework [12]; federated mutual learning [13] distills peer-to-peer. FedDF [4] instead ensembles client models on the server and distills a fused student centrally. Unlike FedDF, FedMKS distributes distillation across clients, removing the server bottleneck at the cost of higher client workload.

C. Statistical Heterogeneity

Zhao et al. [14] quantify the sharp accuracy drop of FL under non-IID data. Weight averaging suffers from client drift, while soft-label distillation provides a weaker per-round signal; our results confirm that KD-only methods degrade more severely under label skew than weight-sharing methods, motivating hybrid designs that combine both channels.

D. Wi-Fi Sensing

Wi-Fi CSI sensing has progressed from coarse localisation to fine-grained activity and gesture recognition [2], [3]. Early systems showed that commodity 802.11 reflections encode

human presence and activities [15], [16]; later work added real-time gesture classification [17], healthcare HAR [18], environment robustness [19], and joint localisation–recognition [20]. Indoor occupancy detection is a representative privacy-critical application: raw CSI reveals presence and behaviour inside a home and cannot be centralised, making it a natural fit for heterogeneous FL.

III. SYSTEM MODEL

A. Notation

B. Network and Data Model

Consider a star-topology FL network with a central server and $K=20$ clients. Each client k holds a private labelled dataset $\mathcal{D}_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k}$, where $x_i^k \in \mathbb{R}^{T \times (4+B)}$ is a preprocessed CSI window and $y_i^k \in \mathcal{Y}_k \subseteq \mathcal{Y}$ its activity label. Under *IID* partitioning each client receives $n_c=30$ examples per class; under *non-IID*, class proportions follow $\text{Dir}(\alpha=0.5)$. All clients also share an unlabelled *public* dataset $\mathcal{D}^{\text{pub}}$ (e.g. HomeHAR) used exclusively for knowledge transfer.

TABLE I
SYMBOL TABLE.

Symbol	Meaning
K	Number of clients ($K=20$)
$\mathcal{K}$	Client index set $\{1, \dots, K\}$
R	Number of FL communication rounds ($R=50$)
r	Round index, $r \in \{1, \dots, R\}$
$\mathcal{D}_k$	Private labelled dataset of client k
n_k	$ \mathcal{D}_k $; $n = \sum_k n_k$
$\mathcal{Y}$	Global label space; $C= \mathcal{Y} $
$\mathcal{Y}_k$	Local label subset, $\mathcal{Y}_k \subseteq \mathcal{Y}$
$\mathcal{D}^{\text{pub}}$	Shared public (unlabelled) dataset
t_k	Tier of client k : S, M, or L
θ_k	All trainable parameters of model f_k
θ_k^{bb}	Backbone parameters (shared by both heads)
$\mathbf{W}_t^{(r)}$	Server-side averaged classifier weights for tier t at round r
d	Distillation-head output dimension (soft-label size)
$s_k^{(r)}$	Soft-label matrix from client k at round r , $s_k^{(r)} \in \mathbb{R}^{ \mathcal{D}^{\text{pub}} \times d}$
$\bar{s}^{(r)}$	Aggregated global soft labels at round r
T, B	CSI window length (1500) and STFT bins (8)

C. Heterogeneous Model Architecture

Each client is assigned a compute tier $t_k \in \{\text{S}, \text{M}, \text{L}\}$ reflecting its hardware budget. Every tier instantiates a *dual-head* model $f_k = (\phi_k, h_k^{\text{clf}}, h_k^{\text{dist}})$, where ϕ_k is a shared backbone parameterised by θ_k^{bb} and the two heads are trained with decoupled losses:

- **Classification head** h_k^{clf} : trained on private data with cross-entropy loss $\mathcal{L}_{\text{CE}}(\theta_k) = -\frac{1}{n_k} \sum_i \log p_k(y_i^k | x_i^k)$.
- **Distillation head** h_k^{dist} : produces a d -dimensional soft-label vector $s_k^{(r)} = h_k^{\text{dist}}(\phi_k(\mathcal{D}^{\text{pub}})) \in \mathbb{R}^{|\mathcal{D}^{\text{pub}}| \times d}$, aligned to the global aggregate via $\mathcal{L}_{\text{KD}}(\theta_k) = \frac{1}{|\mathcal{D}^{\text{pub}}|} \|s_k^{(r)} - \bar{s}^{(r-1)}\|_F^2$.

D. Traditional FL Objective and Its Limitations

FedAvg [1] seeks a single global model minimising the weighted empirical risk:

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_{\text{CE}}^k(\theta), \quad (1)$$

aggregating as $\theta^{(r)} \leftarrow \sum_k (n_k/n) \theta_k^{(r)}$. This requires *identical architectures* across clients; FedProx's proximal term $\frac{\mu}{2} \|\theta_k - \theta^{(r-1)}\|^2$ bounds client drift but keeps the same constraint.

E. Knowledge-Distillation-Based FL Objective

FedMD [11] instead aggregates shared representations: the server maintains global soft labels $\bar{s}^{(r)} = \frac{1}{K} \sum_k s_k^{(r)}$ and each client minimises

$$\mathcal{L}_k(\theta_k) = \underbrace{\mathcal{L}_{\text{CE}}(\theta_k; \mathcal{D}_k)}_{\text{private classification}} + \lambda \underbrace{\mathcal{L}_{\text{KD}}(\theta_k; \mathcal{D}^{\text{pub}}, \bar{s}^{(r-1)})}_{\text{KD alignment}}, \quad (2)$$

where λ balances local accuracy against knowledge absorption. Since $\bar{s}^{(r)}$ lives in a shared d -dimensional space, arbitrary backbones can join the same round.

TABLE II

PER-TIER BACKBONE ARCHITECTURES. EVERY TIER SHARES IDENTICAL INPUT AND OUTPUT SHAPES; ONLY THE BACKBONE DIFFERS.
CONV1D(f, k)/CONV2D(f, k): CONVOLUTION WITH f FILTERS, KERNEL SIZE k ; MP: MAXPOOL(2); GAP: *global average pooling* OVER THE TIME DIMENSION (AVERAGES THE SEQUENCE INTO A SINGLE VECTOR, AVOIDING A LARGE FLATTEN→DENSE WEIGHT MATRIX); BiLSTM(u): BIDIRECTIONAL LSTM WITH u UNITS PER DIRECTION; TB: TRANSFORMER ENCODER BLOCK (4-HEAD MHA, FFN-256, LAYER NORM); DO(p): DROPOUT(p).

Tier	Backbone
<i>CSI input</i> $x \in \mathbb{R}^{1500 \times 12}$ ($T=1500$, $4+B=12$ features)	
S	Conv1D(128,3) → MP → Conv1D(256,3) → MP → Conv1D(256,3) → GAP → Dense(256) → DO(0.30)
M	Conv1D(64,5) → MP → Conv1D(128,3) → MP → BiLSTM(128) → DO(0.30)
L	Conv1D(64,5) → MP → Conv1D(128,3) → MP → BiLSTM(128) → TB → GAP → DO(0.30)
<i>Image input</i> $x \in \mathbb{R}^{28 \times 28 \times 1}$ (MNIST) or $\mathbb{R}^{32 \times 32 \times 3}$ (CIFAR-10)	
S	Conv2D(32,3) → MP → Flatten → Dense(128) → DO(0.25)
M	Conv2D(64,3) → MP → Conv2D(128,3) → MP → Flatten → Dense(256) → DO(0.30)
L	Conv2D(64,3) → MP → Conv2D(128,3) → MP → Conv2D(256,3) → MP → Flatten → Dense(512) → DO(0.40)
<i>Heads (all tiers)</i> : classification Dense(C)+softmax; distillation Dense($d=64$)+ReLU	

F. Problem Definition

Goal. Given K clients with heterogeneous private datasets $\{\mathcal{D}_k\}$ and heterogeneous model tiers $\{t_k\}$, find per-client classifiers $\{f_k^{\text{clf}}\}$ that jointly maximise mean validation accuracy across clients, subject to two constraints:

- 1) *Privacy*: $\mathcal{D}_k$ never leaves client k ; only model parameters and/or soft labels are transmitted.
- 2) *Architectural heterogeneity*: clients in different tiers ($t_j \neq t_k$) instantiate different architectures, so their weight spaces are incompatible and cannot be aggregated directly; cross-tier knowledge transfer must occur in a shared representation space rather than in weight space.

FedMKS (Section IV) meets both constraints by combining per-tier weight averaging (Eq. 1) with cross-tier soft-label distillation (Eq. 2).

IV. METHODS

A. Tiered Dual-Head Architecture

The three tiers use distinct backbones of increasing capacity (Table II). All tiers consume the *same* input shape and expose the *same* two output heads; only the backbone differs. The *classification head* (Dense→Softmax, CE loss) is trained on $\mathcal{D}_k$; the *distillation head* (Dense→ReLU with $d=64$ output units, MSE loss on $\bar{s}$) is trained on $\mathcal{D}^{\text{pub}}$. Gradient flows are fully decoupled between heads; only the backbone is shared.

B. FedMKS: Model and Knowledge Sharing

FedMKS (Fig. 1) addresses the limitations of both FedAvg and FedMD: FedAvg forces every client onto the smallest shared architecture, while FedMD shares only soft labels and forfeits the accuracy gains of weight sharing. FedMKS simultaneously (i) FedAvg's weights within each tier, (ii) distills knowledge across tiers via aggregated soft labels on the public

Algorithm 1 FedMKS — one communication round

Require: Clients $\mathcal{K}$; tiers $t \in \{\text{S,M,L}\}$; $\mathcal{D}^{\text{pub}}$; E_{kd} , E_{loc} ; round r ; current global soft labels $\bar{s}^{(r-1)}$ and tier weights $\{\mathbf{W}_t^{(r-1)}\}$

Ensure: Updated global soft labels $\bar{s}^{(r)}$ and tier weights $\{\mathbf{W}_t^{(r)}\}$

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1: if  $r = 1$  then
2:   Server: send empty payload to each  $k$  ▷ bootstrap
3: else
4:   Server: broadcast  $\bar{s}^{(r-1)}$  and  $\mathbf{W}_{t_k}^{(r-1)}$  to each  $k$ 
5: end if
6: for each client  $k$  in parallel do
7:   if  $r > 1$  and  $\mathbf{W}_{t_k}^{(r-1)}$  received then
8:     Load classifier weights  $\leftarrow \mathbf{W}_{t_k}^{(r-1)}$ 
9:     Receive soft labels  $\bar{s}^{(r-1)}$  as public-set targets
10:    Fine-tune dist. head on  $\mathcal{D}^{\text{pub}}$  for  $E_{\text{kd}}$  epochs minimising
         $\mathcal{L}_{\text{MSE}}(f_k^{\text{dist}}(\cdot), \bar{s}^{(r-1)})$ 
11:   end if
12:   Train clf. head on private data  $\mathcal{D}_k$  for  $E_{\text{loc}}$  epochs minimising
         $\mathcal{L}_{\text{CE}}$ 
13:   Return  $s_k^{(r)} \leftarrow f_k^{\text{dist}}(\mathcal{D}^{\text{pub}})$  and  $w_k^{(r)} \leftarrow \theta_k$ 
14: end for
15: Server:  $\bar{s}^{(r)} \leftarrow \frac{1}{K} \sum_{k=1}^K s_k^{(r)}$ 
16: Server:  $\mathbf{W}_t^{(r)} \leftarrow \text{FedAvg}(\{w_k^{(r)} : t_k=t\})$  for each tier  $t$ 

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TABLE III
ALGORITHM SUMMARY († EFFICIENCY ABLATION ONLY).

Algo.	Soft	Wts	Topology
FedMD	✓	–	Heterogeneous
FedAKD	✓	–	Heterogeneous
FedMKS	✓	✓	Semi-het.
FedDF†	✓	✓	Heterogeneous
FedAvg	–	✓	Homogeneous
FedProx	–	✓	Homogeneous
Local	–	–	Isolated
Central	–	–	Centralised

set, and (iii) applies Mixup augmentation to the public set (following FedAKD [21]).

In round $r=1$ each client trains only on private data and uploads its initial soft labels and classifier weights. From $r=2$ the server returns $\mathbf{W}_{t_k}^{(r-1)}$ and $\bar{s}^{(r-1)}$; the client loads the weights, fine-tunes the distillation head on $\bar{s}^{(r-1)}$ targets, trains the classification head on $\mathcal{D}_k$ for E_{loc} epochs, and returns $s_k^{(r)}$ and $w_k^{(r)}$.

Algorithm 1 formalises one FedMKS round.

Comparison with FedDF. FedDF [4] runs all client models on $\mathcal{D}^{\text{pub}}$ at the server, forms an ensemble $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}; \theta_k)$, and trains a fused student by minimising $D_{\text{KL}}(\bar{s}_\tau \| f_\tau(\mathcal{D}^{\text{pub}}; \theta))$, incurring $O(KP\mathcal{F} + E_{\text{kd}}P(\mathcal{F}+\mathcal{B}))$ operations per round ($P=|\mathcal{D}^{\text{pub}}|$; $\mathcal{F}$, $\mathcal{B}$ = forward/backward cost). FedMKS inverts this: the server only averages tensors, $O(\sum_t |\mathbf{W}_t| + KPd)$, shifting distillation to clients—far better suited to a lightweight edge coordinator.

Table III summarises all evaluated algorithms.

TABLE IV
HYPERPARAMETER SETTINGS (ALL EXPERIMENTS).

Parameter	Value
Rounds R	50
Clients K	20
IID samples / class / client	30 (CSI) / 100 (image)
Non-IID Dirichlet α	0.5
Local epochs E_{loc}	4 (CSI) / 5 (image)
KD epochs E_{kd}	2 (CSI) / 3 (image)
Soft-label dimension d	64
FedProx μ	0.01
Alignment samples	10 000 (cap)
CSI window T	1500
STFT bins B	8

V. EXPERIMENTS AND RESULTS

A. Experiment Design

Datasets. *HomeOccupancy* [22]: 3-class CSI (empty/sleep/work), sessions 1–2 as train, session 3 as test; public data is the *HomeHAR* CSI corpus [22]. *MNIST*: 10-class digit images; public data is CIFAR-10. *CIFAR-10*: 10-class object images; public data is MNIST.

Hyperparameters (Table IV): image benchmarks use $n_c=100$, $E_{\text{loc}}=5$, $E_{\text{kd}}=3$; all other settings are shared. Results are mean validation accuracy over up to 3 seeds; in-progress runs are single-seed (†).

The *main comparison* evaluates all 7 algorithms on 3 datasets under uniform tier heterogeneity (35% S/35% M/30% L), IID and non-IID. FedAvg and FedProx degrade every client to the small model; KD methods and FedMKS keep their assigned tiers. Two ablation studies follow: the first ablates the tier distribution for FedMD on HomeOccupancy (all_small, uniform, skewed=60/30/10); the second compares FedMKS and FedDF on HomeOccupancy (IID, single seed) on accuracy, server/client compute, and communication bytes.

B. Results

1) *Algorithm Comparison:* Figs. 3 and 4 show validation accuracy over 50 rounds under IID and non-IID partitions.

HomeOccupancy. Under IID, FedMKS (90.4%) and FedAvg (89.5%) approach the centralised ceiling (90.8%), while FedMD (88.5%) and FedAKD (87.0%) lag without weight sharing; Local plateaus at 80.4%. Under non-IID, completed weight-sharing runs (FedAvg 38.4%, FedProx 33.7%) already trail local-only training (60.1%).

MNIST. IID: Central 99.0%, FedMKS 97.7%, FedAvg/FedProx 97.0%/97.2%, FedMD/FedAKD/Local $\approx 95.8\%$. Non-IID: FedAvg 97.4%, FedProx 97.2%, FedMKS 96.6%, while FedMD/FedAKD/Local plateau at $\approx 84\%$ — ≈ 13 points below the all-small weight-sharing methods despite their larger architectures.

CIFAR-10. Cross-domain public data (MNIST) gives little signal: all FL methods fall 10–30% below centralised. FedAvg (53.8%) edges FedProx (52.9%) and FedMKS (52.6%) under

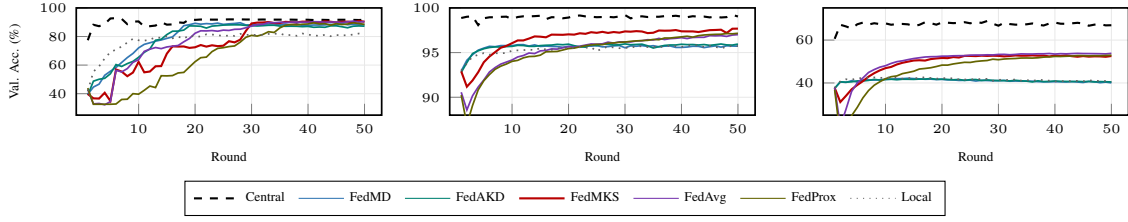


Fig. 3. Validation accuracy over 50 rounds under IID partitioning (mean over completed seeds). Panels: HomeOccupancy, MNIST, CIFAR-10. FedAvg/FedProx use the all-small architecture. † single-seed runs.

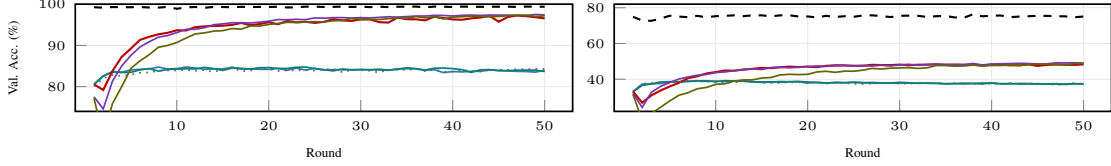


Fig. 4. Validation accuracy under non-IID partitioning (Dirichlet $\alpha=0.5$). Panels: MNIST, CIFAR-10 (HomeOccupancy non-IID runs in progress).

IID, and leads under non-IID (49.1% vs. 48.9%/48.3%). KD-only methods (FedMD 37.2%, FedAKD 37.4%) do not beat Local (37.3%).

a) Key observation.: Under non-IID, weight sharing beats cross-architecture distillation even when weight sharers are forced onto the smallest model (MNIST: $\approx 97\%$ vs. $\approx 84\%$). FedMKS is designed to recover model-sharing accuracy while preserving the heterogeneity that only distillation permits.

2) Ablation Studies:

a) Tier heterogeneity.: Fig. 5 ablates the tier distribution for FedMD on HomeOccupancy. Under IID all configurations converge to $\approx 86\%$; under non-IID uniform (61.5%) beats *all_small* (59.1%) and skewed (58.6%)—small-dominated tiers underperform a balanced mix under label skew.

b) FedMKS vs. FedDF.: Fig. 6 and Table V compare FedMKS and FedDF on HomeOccupancy (IID, single seed, 30 rounds, 10 clients). FedDF converges faster (plateau near round 10, 89.0% vs. 87.1% final-five mean), though FedMKS is still climbing and edges ahead at round 30 (88.2% vs. 88.1%). The efficiency profiles differ sharply: FedDF spends ≈ 243 s/round on server-side ensemble training, whereas FedMKS does essentially no server compute (≈ 0.06 s) and distributes distillation across parallel clients (≈ 93 s vs. ≈ 25 s per client). FedMKS removes the serial server bottleneck at $\approx 22\%$ higher communication (37.9 vs. 31.0 MB/round); the whole ≈ 6.9 MB overhead is the soft-label exchange ($|\mathcal{D}^{\text{pub}}| \times d$ each way), which FedDF avoids by distilling on the server. The dashed curve in Fig. 6 shows FedMKS with Mixup disabled on the public set: convergence is comparable and the final accuracy drops only ≈ 1 pp (87.3% vs. 88.2% at round 30), indicating that Mixup yields a modest regularisation benefit rather than being essential to the hybrid update.

TABLE V
EFFICIENCY COMPARISON ON HOME OCCUPANCY (PER ROUND UNLESS NOTED; ACCURACY = FINAL-FIVE-ROUND MEAN).

Metric (per round)	FedMKS	FedDF
Final val. accuracy (%)	87.1	89.0
Server compute (s)	0.06	243.2
Mean client compute (s)	93.1	24.6
Comm.: model weights (MB)	31.0	31.0
Comm.: soft labels (MB)	6.9	0.0
Communication total (MB)	37.9	31.0
Total communication (MB, 30 r.)	1137.7	930.1

VI. CONCLUSION

We presented **FedMKS**, a heterogeneous federated knowledge-distillation algorithm for CSI-based indoor sensing that couples intra-tier weight sharing with cross-tier soft-label distillation. FedMKS reaches near-centralised accuracy on HomeOccupancy ($\Delta \approx 0.5\%$ under IID) while preserving architectural heterogeneity, and matches FedDF accuracy (87.1% vs. 89.0%) while eliminating its ≈ 243 s/round server bottleneck (≈ 0.06 s/round) at $\approx 22\%$ higher communication.

VII. LIMITATIONS AND FUTURE WORK

The main comparison uses a single uniform tier distribution; FedMKS also incurs higher communication than FedMD, motivating weight compression (quantisation, top- k). Cross-domain public data gave little signal on CIFAR-10, so domain-adapted or synthetic public sets remain open. Finally, HomeSenseKD assumes synchronous rounds; asynchronous aggregation under dropouts is an important extension.

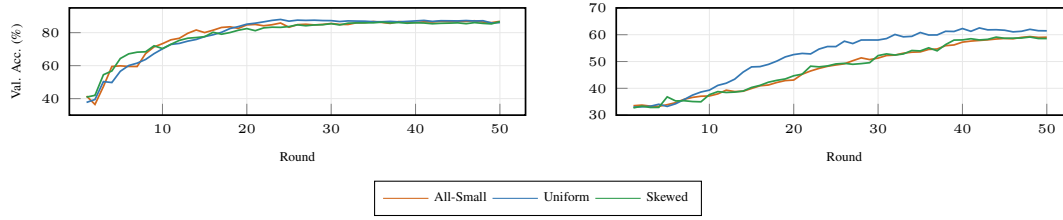


Fig. 5. FedMD tier-heterogeneity ablation on HomeOccupancy, IID (left) and non-IID (right): all_small (100% S), uniform (35/35/30), skewed (60/30/10).

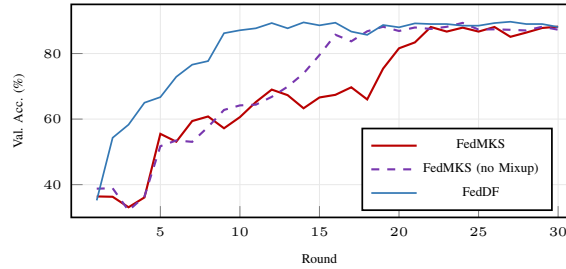


Fig. 6. FedMKS vs. FedDF validation accuracy on HomeOccupancy (IID, 30 rounds, 10 clients). The dashed curve is a FedMKS ablation with Mixup augmentation on the public set disabled.

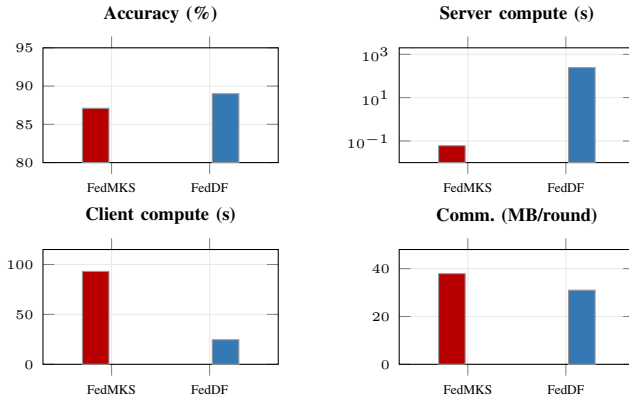


Fig. 7. Compute, communication, and accuracy comparison of FedMKS and FedDF on HomeOccupancy (cf. Table V). FedMKS eliminates the server bottleneck at $\approx 22\%$ higher communication and higher per-client compute.

CODE AND DATA AVAILABILITY

Source code is available at <https://github.com/gadm21/HomeSenseKD>. HomeOccupancy and HomeHAR datasets are hosted on HuggingFace at <https://huggingface.co/gadgadgad>.

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